MATH9102: Fundamentals of Data Analysis

Unit 6: Linear Regression

One Model for Prediction and for Difference

Today

1. **What Regression Does** — and what it can claim
2. **One Predictor** — a line through a cloud
3. **How Good Is the Model?** — F, R^2, coefficients
4. **Several Predictors** — a dummy, two parallel lines
5. **When the Lines Need Not Be Parallel**
6. **Three or More Categories**
7. **Pupils in Schools** — clustering and precision

One dataset all session; four models, each changing one thing.

Important Note

A coefficient describes an association in these data. Whether it is a cause is a question about the design, and the model cannot answer it.

What Regression Does

Where This Week Sits

Units 3 to 5 asked two kinds of question. Regression asks both at once.

What you have

- Unit 3: how strongly two measurements move together
- Units 4 and 5: whether groups differ on an outcome
- An effect size beside every test, in words

What is new

- One model that predicts *and* compares groups
- A prediction, and what it does not tell you
- Several predictors at once, each adjusted for the others
- A category entered as a dummy variable

Continuous outcome, any predictors

The outcome is a measurement. The predictors can be measurements, categories, or both. Unit 9 adapts the model to a yes-or-no outcome.

What a Regression Model Is

An outcome, written as a weighted sum of predictors, plus what the sum misses.

$$y = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k + e$$

The parts

- y — the outcome; $x_1, \ldots, x_k$ — the predictors
- b_0 — the intercept; $b_1, \ldots, b_k$ — coefficients
- e — the residual: observed minus predicted

How it is fitted

By least squares: the line, or plane, closest to the data. Then it predicts cases it has not seen.

A reason for every predictor

Each predictor needs a substantive reason before it goes in. One added because it was in the file is a fishing trip.

Prediction and Difference, One Model

The same equation answers two kinds of question. What a coefficient means depends on the predictor.

A continuous predictor gives a rate

- The change in predicted y for one unit more of x
- Reading score at 11 $\rightarrow$ exam score at 16

A categorical predictor gives a difference

- The gap between a group and the reference group
- Girls against boys; one school type against another

Questions it answers

- How well does a set of predictors predict the outcome?
- Does a predictor still predict once the others are held fixed?
- How large is a group difference, after accounting for the other predictors?

What Regression Can and Cannot Claim

A model finds association. A causal claim needs three more things, and none of them is in the output.

Association — the variables move together: higher values of one go with higher (or lower) values of the other

Time order — the cause comes first. Necessary, and nowhere near sufficient

Non-spuriousness — no third variable drives both. Ice-cream sales and drownings rise together; hot weather drives both

Direction — the model does not know which way it runs. Stress may lower perceived control, or the reverse

Say "is associated with", not "causes"

The design and your knowledge of the subject decide whether a cause is plausible; the model never does. Unit 3's *Correlation Is Not Causation* applies to every coefficient you fit today.

One Predictor

The Data: Pupils in Schools

4,059 pupils aged 16, in 65 London secondary schools.

What was measured

- `normexam` — exam score at 16, normalised
- `standlrt` — London Reading Test at 11, standardised
- `girl` — sex, held as text: boy or girl
- schgend — mixed, boys' or girls' school
- `school` — which school

Where it comes from

- A subset of the exam data Goldstein and colleagues analysed: 5,748 pupils in 66 schools in six Inner London Education Authorities
- Distributed with MLwiN, to teach *multilevel* modelling — which matters at the end of the session

Goldstein, H., Rasbash, J., Yang, M., Woodhouse, G., Pan, H., Nuttall, D. and Thomas, S. (1993). A multilevel analysis of school examination results. *Oxford Review of Education*, 19(4), 425–433.

Model 1: Reading at 11, Exams at 16

Hypotheses

H_0: Reading score at 11 does not predict exam score at 16 — the slope is zero.

H_1: It does — the slope is not zero.

Both variables are standardised

- Mean near zero, SD near one: a score is a distance from the average, in SDs
- So $x = 0$ is the average reader, and the intercept is the prediction for them
- A convenience for reading the model, not a requirement of it

Watch out

That reading of zero holds for a continuous predictor. A categorical predictor's zero is a *category* — in Model 2, the boys.

Stand Up — 5 minutes

Five minutes. The question is on the table; the data are not.

Food for thought

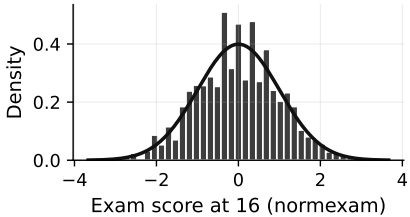
- Will reading at 11 predict exams at 16? Which way?
- A pupil one SD above average at 11: more, less, or exactly one SD above average at 16?
- Of all the spread in exam scores, how much will reading account for — a tenth, a third, most of it?

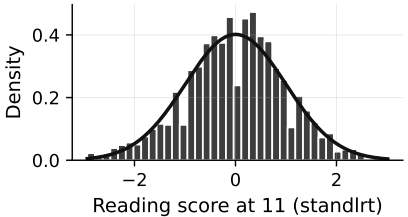
Say it out loud

Tell the person next to you, or jot it on paper — not on a screen. A prediction made before the output is a prediction. One made afterwards is a story.

Look Before You Model

Both variables, on their own, before any model.





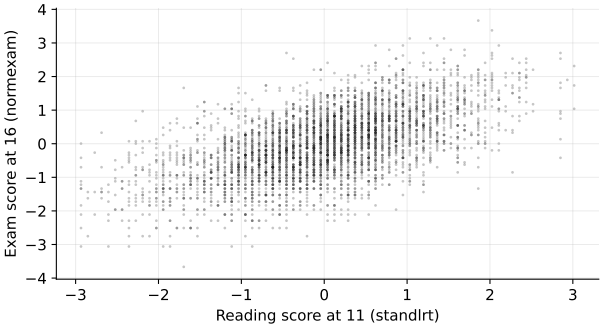
What you are looking for

- Floors, ceilings, gaps
- Impossible values, wild outliers

Watch out

Not normality of the outcome. Regression's assumptions are about the residuals, which are Unit 7's.

Scatter, Then r



One cloud, rising to the right, roughly straight.

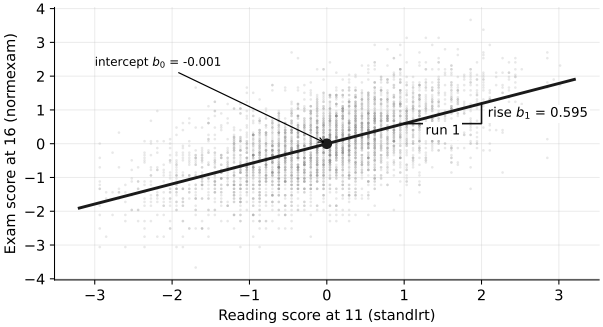
$r = .59$ — large against Unit 3's benchmarks.

One verdict, one benchmark

Describe a coefficient once, against a stated benchmark. The same r cannot be "strong" on one page and "moderately strong" on the next.

A straight line is a reasonable summary of this cloud. That is the condition for fitting one: look first.

A Line Through a Cloud



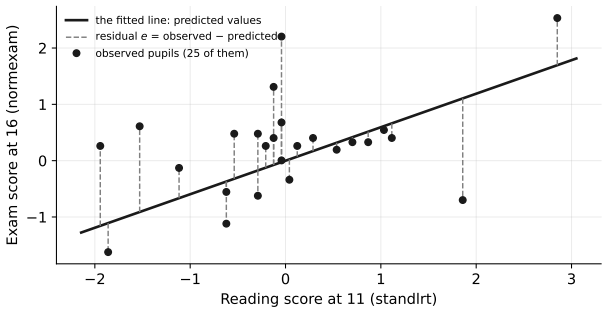
$$\hat{y} = b_0 + b_1 x$$

- $\hat{y}$ — the *predicted* outcome
- b_0 — the intercept: $\hat{y}$ when $x = 0$
- b_1 — the slope: the change in $\hat{y}$ for one unit more of x

One line among many

Every line through the cloud makes a prediction for every pupil. Least squares picks one — the next slide says which.

Residuals and Least Squares



The residual is observed minus predicted.

$$e_i = y_i - \hat{y}_i$$

Least squares makes $\sum e_i^2$ as small as possible.

- Squared, so misses above and below do not cancel
- And a miss twice as large counts four times as much

A residual is not a mistake

It is what the model did not capture about that pupil: everything that shapes an exam score apart from reading at 11.

Fitting It

```
import statsmodels.formula.api as smf

regression = m9.load_regression()

# Outcome on the left of the tilde, predictor on the right.
model1 = smf.ols("normexam ~ standlrt", data=regression).fit()
print(model1.summary())
```

The formula

- Outcome ~ predictors, as text
- Columns named, data passed by name
- The intercept is included without asking

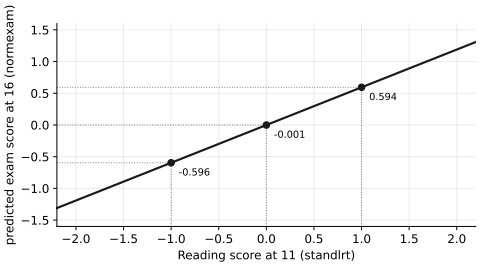
Read the summary once

Then take the pieces you need off the fit: `params`, `rsquared`, `fvalue`, `conf_int()`. The next slides take them one at a time.

Using the Model to Predict

Predicted exam score = $-0.001 + 0.595 \times$ reading score

```
# Predictions for three new pupils: one SD below the mean, at it, above it.
new_pupils = pd.DataFrame({"standlrt": [-1, 0, 1]})
model1.predict(new_pupils)
```



One SD below average at 11: -0.596; at the average: -0.001; one SD above: 0.594. Each step of one SD adds the slope, 0.595.

Watch out

A prediction is the expected score for pupils with that reading score. Any one of them can land well above or below it.

How Good Is the Model?

How Good Is the Model? The Mean as a Baseline

With no model at all, your best guess for every pupil is the mean.

The baseline

- Predict the mean exam score for everyone
- Wrong for nearly everyone, by however far each pupil sits from the mean
- That total error is what a model has to beat

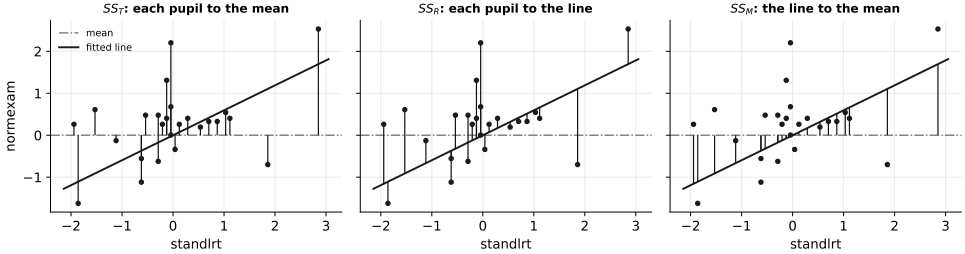
Three questions

- Does the line do better than the mean?
- By how much?
- Is that more than chance would give?

Every measure of fit compares the line with the mean

F asks whether the improvement is real; R^2 says how big it is. Both come from the same three sums of squares.

Three Sums of Squares



- SS_T — total: every pupil from the mean
- SS_R — residual: every pupil from the line; what is left
- SS_M — model: the line from the mean; what the line gained

They add up

$SS_T = SS_M + SS_R$

Model 1: $4049.4 = 1417.5 + 2631.9$, over all pupils. The picture shows twenty-five of them.

The F-Test of the Whole Model

```
import statsmodels.api as sm

# The sums of squares, the mean squares and F, for the whole model.
sm.stats.anova_lm(model1)

# Or straight off the fit: F with both degrees of freedom, and R squared.
model1.fvalue, model1.df_model, model1.df_resid, model1.rsquared
```

As in Unit 5: signal over noise

$$F = \frac{MS_M}{MS_R} = \frac{SS_M/df_M}{SS_R/df_R}$$

df_M is the number of predictors; df_R is n minus that, minus one.

Model 1

$F(1, 4057) = 2185.01$, $p < .001$

H_0: the model predicts no better than the mean. With one predictor, F is the slope's t squared.

The share of the total variation that the model accounts for.

$$R^2 = \frac{SS_M}{SS_T}$$

Model 1

- $R^2 = .350$: reading at 11 accounts for about 35% of the variation in exam scores at 16, in this sample
- With one predictor, R^2 is r^2: .592 squared
- The rest is everything the line leaves out

Watch out

"Accounts for" describes fit in this sample. It is not a measure of how important reading is, and not a share of the cause. The last section comes back to this.

Reading the Coefficients

Term	b	What it says
Intercept	-0.001	The predicted exam score for a pupil at `standlrt` $= 0$, the average reader. Close to zero, because both variables are standardised
`standlrt`	0.595	For each SD more reading at 11, the predicted exam score at 16 is 0.595 SD higher

Each coefficient has a test

- H_0: $b = 0$, a t-test on $n - k - 1$ df
- For the slope, it is the F-test's question again

Sign, then size, then units

The sign gives the direction. The size gives the rate — in the units of the variables, which here are SDs.

Coefficients Come With Intervals

Slope 0.595, 95% CI [0.570, 0.620].

How to read it

- The range of slopes this sample is compatible with
- Narrow, because four thousand pupils pin a slope down well
- It excludes zero exactly when the slope's test is significant at .05

Why report it

- A *p*-value speaks only to whether the slope could be zero
- The interval says how large it plausibly is — the question anyone using the result actually has

Watch out

The interval assumes the pupils are independent of each other. These pupils share schools. The last section shows what that does to it.

Standardised Coefficients

```
# Standardise inside the formula: both variables in SD units.
beta = smf.ols(
    "standardize(normexam) ~ standardize(standlrt)",
    data=regression,
).fit()
beta.params
```

Every variable in SDs

- $\beta = 0.592$: one SD more reading, 0.592 SD more exam score
- With one predictor, β is r
- Useful when predictors come in different units; here both are in SDs already, near enough, so β and b nearly agree

Watch out

Never standardise a binary predictor. "One SD more girl" means nothing: report its unstandardised b, which is a difference between two groups.

Reporting Model 1

```
# Every number in the paragraph comes from the fitted model.
m9.report_regression(
    model1,
    outcome_label="exam score at 16",
    predictor_labels={"standlrt": "reading score at 11"},
)
```

The paragraph carries

- F with **both** degrees of freedom, and p
- R^2, and what it accounts for
- The fitted equation
- Each coefficient with its interval, in words
- What the model does not establish

Nothing was held constant

"Holding other factors constant" belongs to a model with other factors in it. With one predictor there is nothing to hold, and the phrase claims a control the model never applied.

Break — 30 minutes: Fit Model 1

20 minutes on the analysis, then 10 for tea.

In `notebook.ipynb`

- **Part 1**, top to bottom
- Read each output before running the next cell

Then in `exercise.ipynb`

- Question 1 — sleepiness predicted from anxiety, in the sleep study

Think about

Compare Model 1 with the prediction you made at the stand-up. Which did you misjudge — the direction, the slope, or the share of the variation?

Then stop

The last ten minutes are tea, not catch-up. If Part 1 did not finish, bring it to the end of the session.

Where You Should Be: Model 1

You should now have

- Model 1 fitted with the formula, and its summary read once
- The slope, its interval and R^2, each said in a sentence
- A generated paragraph, with no number typed into it

The two that catch people

Treating the intercept's p-value as a finding. And reading R^2 as how much of exam success reading *causes*.

Several Predictors

Several Predictors

One outcome, several predictors, one coefficient each — and each coefficient adjusted for all the others.

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$

What b_j now means

- The change in $\hat{y}$ for one unit more of x_j
- . . . among cases with the same values of every other predictor
- With two predictors the line becomes a plane

"Holding the others fixed"

Is a statement about the model's arithmetic, not about an experiment. Nobody held anything fixed: the model compares cases that differ on one predictor and match on the rest — as far as a straight-line model can.

Categorical Predictors Become Dummies

A category has no units, so "one unit more" means nothing. It enters as an indicator instead.

A dummy variable

- 1 for the category of interest, 0 for the **reference**
- Its coefficient: the gap between that category and the reference, at the same values of the other predictors
- The formula builds it from a text column, and picks the reference alphabetically unless you name one

No test first

You do not need a *t*-test to "establish" a difference before a group goes into a model. Describe the groups, then let the model estimate the gap — adjusted for the other predictors, which a *t*-test cannot do.

Model 2: Does Sex Add Anything?

Hypotheses

H_0: Among pupils with the same reading score at 11, girls and boys have the same mean exam score at 16.

H_1: They do not.

Why adjust for reading

- If girls and boys arrived at 11 reading at different levels, a raw comparison at 16 mixes two things
- The model separates them: the gap among pupils who read alike

What changes

- One dummy, `girl`; boys are the reference
- Model 1 sits inside Model 2: drop the dummy and you have it back

Model 2

```
# girl holds text, "boy" or "girl": the formula builds the dummy.
# The reference is the first level alphabetically, so boys.
model2 = smf.ols("normexam ~ standlrt + girl", data=regression).fit()
model2.params
```

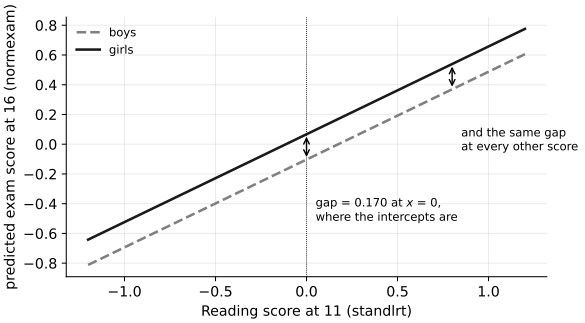
Predicted exam score $= -0.103 + 0.591 \times$ reading score $+ 0.170 \times$ girl

- `girl` is 1 for girls and 0 for boys
- Among pupils who read alike at 11, girls' predicted scores are 0.170 SD higher; $p < .001$

The reading slope moved

From 0.595 to 0.591. Adding a predictor changes the others' coefficients whenever the predictors are related. Each is now adjusted for the rest.

Two Parallel Lines



Boys (girl $= 0$):

$-0.103 + 0.591 \times$ reading

Girls (girl $= 1$):

$0.067 + 0.591 \times$ reading

The same slope; intercepts 0.170 apart.

Say where the gap is read

The intercepts are the predictions at reading score zero, the average reader. Put in a reading score of one and you are describing pupils one SD above average, not "the average student". With parallel lines the gap is the same everywhere — so say that, or say where you read it.

The Intercept

The prediction when every predictor is zero. In Model 2: a boy (girl $= 0$) who read at the average at 11 (`standlrt` $= 0$).

When it means something

- Predictors are centred or standardised
- Zero on every predictor is a case that exists

When it does not

- Zero lies outside the data: age zero, no income at all
- Then the intercept only anchors the line

Watch out

The intercept's test asks whether that one prediction is zero — rarely a question anyone asked. A significant intercept is not a finding.

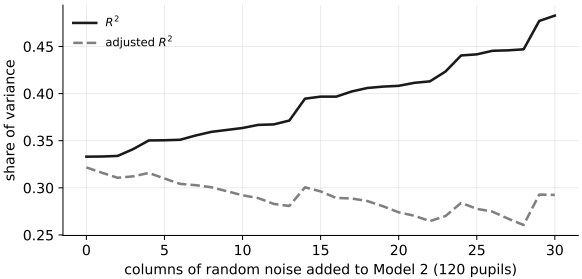
Four Things to Read

Statistic	The question it answers
F and its p	Does the model as a whole predict better than the mean?
R^2, adjusted R^2	How much of the variation does it account for?
Each coefficient b	Which way, and how steeply, adjusted for the others?
Each b's test and interval	Is it distinguishable from zero, and how precisely is it known?

In that order, and all four

A significant F says something in the model predicts. Only the coefficients say what; only the intervals say how precisely; and only R^2 says whether any of it predicts much.

R^2 Never Falls: Adjusted R^2



Add any predictor — even random noise — and R^2 goes up.

Adjusted R^2 charges for each one:

$$1 - (1 - R^2)\,\frac{n-1}{n-k-1}$$

It still rises whenever a new predictor's $|t|$ exceeds one, so even noise lifts it now and then.

One quantity, one precision

Model 2: $R^2 = .357$, adjusted .357. Say which you report, then quote it to the same precision everywhere — a number given two ways in one report reads as two numbers.

Reporting Model 2, and Comparing It With Model 1

```
# Compared models must be fitted on the same rows: select every
# variable either model uses, drop missing rows once, fit both on that.
assert model1.model.data.row_labels.equals(model2.model.data.row_labels)

# Does girl add anything? The F-test for the extra term.
sm.stats.anova_lm(model1, model2)

# And how much R squared moved.
model2.rsquared - model1.rsquared
```

What `girl` adds

- $F(1, 4056) = 43.70$, $p < .001$
- R^2 rises by .007
- Clearly not zero, but negligible by Cohen's convention: a gap of 0.170 SD

Same rows, or no comparison

Models fitted on different rows differ in their data as well as their predictors, so their F and R^2 cannot be compared. Select every variable any model uses, drop missing rows once, fit them all on that.

When the Lines Need Not Be Parallel

When the Lines Need Not Be Parallel

Model 2 forces girls and boys onto the same slope. Should it?

Hypotheses

H_0: Reading at 11 predicts exams at 16 equally steeply for girls and boys.

H_1: The slopes differ.

An interaction

- The slope of one predictor depends on the value of another
- The two do not simply add

$$\hat{y} = b_0 + b_1 x + b_2 g + b_3 (x \times g)$$

b_3 is the difference between the girls' slope and the boys'.

Stand Up — 5 minutes

Five minutes. A second question, and again no data.

Food for thought

- Is the gap between girls and boys wider among strong readers, among weak readers, or the same?
- How different would the slopes have to be before you cared?
- If the interaction is not significant, what may you say — and what may you not?

Say it out loud

Write the prediction down before the next slide. The output will show the answer, and you will want to know whether you saw it coming.

Building the Interaction in the Formula

```
# standlrt * girl means standlrt + girl + standlrt:girl.
# The product is built by the formula, so it always matches its parts.
model3 = smf.ols("normexam ~ standlrt * girl", data=regression).fit()
model3.params
model3.pvalues["standlrt:girl[T.girl]"]
```

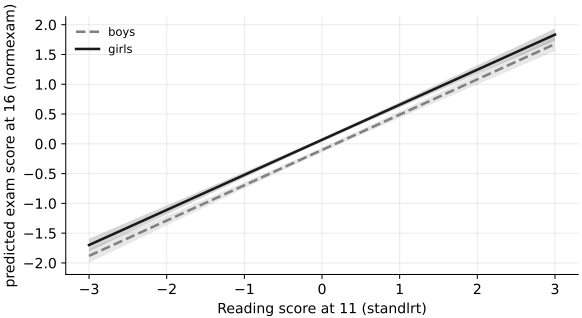
What the star does

- `a * b` is `a + b + a:b`: both main effects and their product
- The product is computed from the columns each time the model is fitted

Watch out

A product column stored in the data is the same thing while it is right, and silently wrong the day either part is recoded. Build it in the formula.

Model 3



Two slopes, barely different.

- Boys' slope: 0.593
- Girls' slope: 0.589
- Their difference, b_3: -0.004
- Adjusted R^2: .357, against Model 2's .357

Read the picture, then the coefficient

The lines stay almost the same distance apart: that gap is the girls' advantage, as in Model 2. The bands show where each line lies, not whether the slopes differ. That is b_3, and its interval is on the next slide.

Reporting a Non-Significant Interaction

Interaction $b_3 = -0.004$, 95% CI $[-0.054, 0.046]$, $p = .874$.

What it says

- No evidence that the slopes differ
- Any difference is small beside the slopes themselves: the interval bounds it
- So keep Model 2, the simpler model

What it does not say

- That the slopes are equal
- That sex is unrelated to exam scores — that is the main effect, in Model 2

One coefficient, one p

Generate the sentence from the fit and it cannot quote two p-values for one coefficient. And "moderates" or "differs across" needs this model: an additive model has no term that could show it.

Three or More Categories

k Categories, $k - 1$ Dummies

School type has three levels: mixed, boys', girls'. Two dummies carry it; the third level is the reference.

- Each dummy's coefficient is a difference from the reference
- Pick the reference on purpose: the comparison you want, and a group large enough to compare with
- A third dummy would be redundant: the three always sum to one

Check the cells

A cross-tabulation shows which combinations of predictors exist. It takes one line, and the next slide is why it matters.

```
# Before any model: who attends which type of school?
pd.crosstab(regression["schgend"], regression["girl"])
```

Model 4: Check the Combinations First

Model 4 adds school type to Model 2. Before reading anything as "holding sex fixed", count who attends which school.

	Boys	Girls
Mixed school	1,110	1,059
Boys' school	513	**0**
Girls' school	**0**	1,377

No girl attends a boys' school, and no boy a girls' school.

So "a boys' school, holding sex fixed" cannot be about a girl: there are none to hold. Within a single-sex school sex never varies, so girls and boys are compared only in the mixed schools.

Before any "holding fixed", check the combinations exist

The cross-tab says which comparisons the data contain. The model output never will: it prints a coefficient either way.

School Type, Read Within Sex

```
# School type beside sex, mixed schools the reference. Each school-type
# coefficient compares one sex with the same sex at mixed schools.
model4 = smf.ols(
    "normexam ~ standlrt + girl + C(schgend, Treatment('mixedsch'))",
    data=regression,
).fit()
model4.params
```

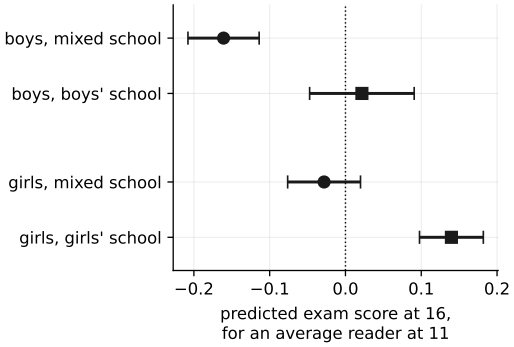
What each coefficient compares, at the same reading score

- `girl`: girls against boys, in mixed schools: 0.133
- Boys' school: boys there against boys at mixed schools: 0.183
- Girls' school: girls there against girls at mixed schools: 0.168

Within sex, and not an effect of

Each school-type coefficient compares one sex with the same sex. A prediction for a girl at a boys' school is about nobody: do not report one. And none of it measures what a school does: families choose schools.

Reporting Model 4



At the same reading score at 11, boys at boys' schools scored 0.183 SD higher at 16 than boys at mixed schools (95% CI [0.099, 0.266], $p < .001$), and girls at girls' schools 0.168 SD higher than girls at mixed schools ([0.104, 0.232], $p < .001$). In mixed schools, girls scored 0.133 SD higher than boys ($p < .001$). The model as a whole: $F(4, 4054) = 580.15$, adjusted $R^2 = .363$.

Watch out

Every interval on this slide treats 4,059 pupils as independent of one another. They sit in 65 schools. The next section asks whether that matters.

Break — 30 minutes: Dummies, Lines and Interactions

25 minutes on the analysis, then 5 to stretch.

In `notebook.ipynb`

- **Parts 2, 3 and 4** — Model 2 against Model 1, the interaction, and school type

Then in `exercise.ipynb`

- Questions 2 to 4 — trouble falling asleep, caffeine and alcohol together, and education level

Think about

In Question 4, which education level should be the reference — and how many people are in it? Count before you choose.

Then stop

Five minutes on your feet before the last section. It is short, and it changes how you read everything above.

Where You Should Be: Several Predictors

You should now have

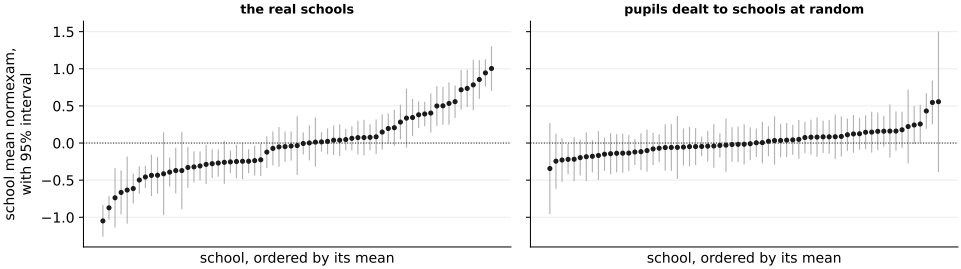
- Model 2 compared with Model 1, both fitted on the same rows
- An interaction built in the formula, reported with its interval
- A three-level factor with its reference named, and its cells counted

The two that catch people

Comparing models fitted on different rows. And a reference category with a handful of people in it, which makes every other level look different.

Pupils in Schools

Pupils in Schools: The ICC



Intraclass correlation: ICC $= .153$ — the share of the variation in exam scores that lies *between* schools.

The design effect: $1 + (m - 1) \times \text{ICC} = 10.4$, with 62.4 pupils in the average school.

What it means

Pupils in one school are more alike than pupils drawn at random. For the average exam score, 4,059 of them carry about as much information as 391 unrelated pupils would.

What Clustering Does to Standard Errors

```
# The same model, with standard errors that let pupils in one school be alike.
model4_cl = smf.ols(
    "normexam ~ standlrt + girl + C(schgend, Treatment('mixedsch'))",
    data=regression,
).fit(
    cov_type="cluster", cov_kwds={"groups": regression["school"]}, use_t=True,
)

# How much wider each standard error is. The coefficients do not move.
model4_cl.bse / model4.bse
```

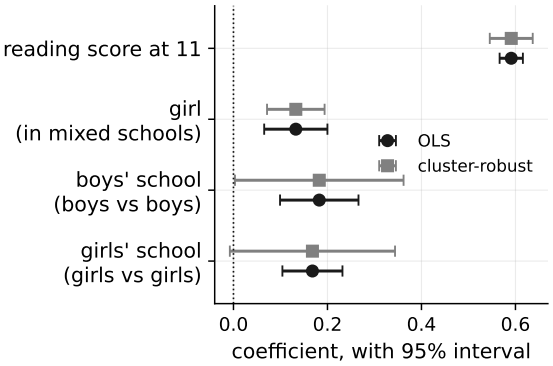
Cluster-robust standard errors

- Let pupils in one school be alike; assume schools are independent
- The coefficients do not move — only their standard errors do

How much wider, for Model 4

Reading at 11: $\times 1.81$
Girl, in mixed schools: $\times 0.89$
Boys' school: $\times 2.11$
Girls' school: $\times 2.70$

Clustering Changes a Conclusion



School type loses most: it varies only between schools — 10 boys' and 20 girls'.

Boys' schools: $[0.003, 0.362]$, $p = .046$
Girls' schools: $[-0.008, 0.344]$, $p = .061$

The same size, both now reaching to about zero. One p falls each side of .05: read the intervals, not the side.

A repair, not the treatment

The model for pupils in schools is multilevel, beyond this module. You can always notice the clustering, and say what it does to your claims.

Before You Write It Up

Two Conclusions You Must Not Draw

A significant coefficient

Does *not* mean the predictor causes the outcome. Pupils who read well at 11 do better at 16; the model cannot say that reading is why. Prior attainment travels with family, school and much else that was never measured.

"Reading accounts for 35% of exam performance"

Does *not* make reading 35% of the reason. R^2 describes how closely a line fits these pupils, given how spread out this sample is. A sample with a narrower range of readers would give a smaller R^2 for the same slope.

Scope

You SHOULD be able to

- Fit simple and multiple regressions with the formula
- Read F, R^2 and each coefficient with its interval, in words
- Predict from the fitted equation, and say what a prediction is
- Enter a category as dummies with a named reference, and check that the combinations exist
- Build an interaction in the formula; report a non-significant one honestly
- Compare nested models fitted on the same rows
- Say what clustering does to standard errors, and what it leaves alone

This week: the notebook fits all four models on the exam data; the exercise asks you to do the same on the sleep study.

That's all for this week